Entity-Token World Models and Differentiable Predictive Control for Building-Agnostic HVAC Control

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Abstract

A single controller for the Heating, Ventilation and Air Conditioning (HVAC) systems of unseen buildings has to cope with different numbers of zones and air handlers, different thermal properties and climates ranging from subarctic winters to tropical summers. This work addresses the Smart Buildings Challenge by representing every zone, air handler and hot-water system as a token of the same transformer architecture, used both for a world model of the building and for the policy, and by training the policy with Differentiable Predictive Control (DPC), that is, by backpropagating the competition reward through closed-loop rollouts of up to 48 hours inside world models refined in on-policy rounds. On the 30 official scenarios over 14 days, the score of the Proportional-Integral-Derivative (PID) baseline falls from 0.0591 to 0.0103, mostly through a lower discomfort, and an earlier policy reaches 0.013 on the Kaggle public leaderboard. The analysis shows that the remaining loss is concentrated in the subarctic climate, that the differences between the most accurate world models are smaller than the spread between training seeds, and that a receding-horizon search with the true simulator still recovers about 30% of the loss in a cold and a mild scenario.

1 Introduction

Heating, Ventilation and Air Conditioning (HVAC) systems are still mostly operated by rule-based logic tuned by hand for each building [4,13], since Model Predictive Control (MPC) and reinforcement learning, despite large savings, require a model or a training process specific to every building [11]. The Smart Buildings Challenge [7], built on the Smart Buildings Control Suite simulator [6], makes this obstacle explicit by asking for one policy that controls buildings never seen during development, scored by a reward that combines occupant comfort, energy cost and carbon emissions.

The approach presented here describes the building as a set of entities (zones, air handlers, the hot-water system and a global token), processed by the same transformer architecture in a world model and in the policy, so that both apply to any building, and trains the policy, neither by trial and error nor by imitation, by Differentiable Predictive Control (DPC) [5], maximizing the competition reward rebuilt from the world-model predictions through closed-loop rollouts, as outlined in Figure 1. The main contributions of this work are the following:

- an entity-token world model with long thermal memory, refined in on-policy rounds, with a closed-loop temperature error of 0.79 K after 24 hours on held-out buildings;
- a DPC scheme with rollouts of up to 48 hours, a command filter inside them and two alternated world models, which reaches 0.0103 on the 30 official scenarios, against 0.0591 for the PID baseline;

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- an analysis of where the gains come from, by reward term, climate and control behavior, of the calibration of the world models and of the spread between training seeds, together with an oracle bound on the remaining headroom.

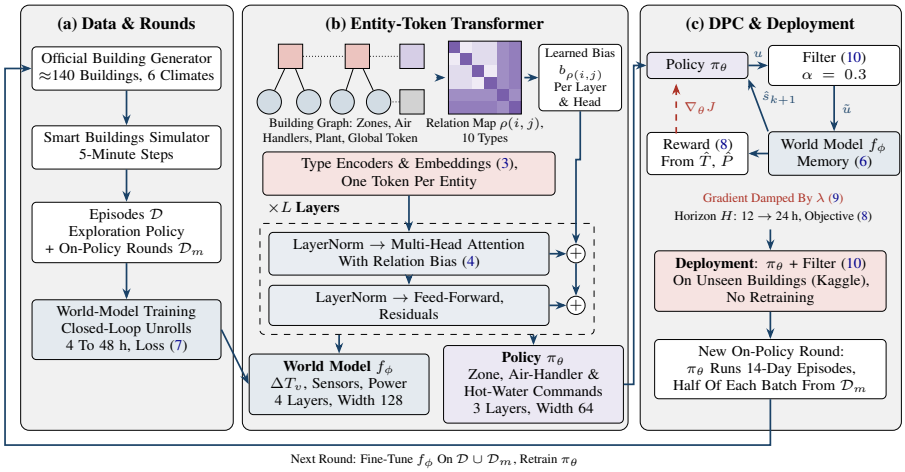


Figure 1: Overview of the method: (a) data collection on generated buildings and on-policy rounds; (b) entity-token transformer shared by the world model and the policy (same structure, separate weights), with Graphormer-style attention biases given by the relation between two entities; (c) DPC training of the policy through the world model and deployment on unseen buildings.

2 Related Works

Unlike the detailed individual test cases of the Building Optimization Testing (BOPTEST) framework [2], the Smart Buildings Challenge evaluates a single policy on buildings generated at scale. MPC is the reference for advanced building control [4], but requires a model identified for every building. Deep reinforcement learning has been applied to HVAC control since [14], with sample efficiency, safety and transfer to new buildings as its main obstacles [11,13].

Differentiable MPC [1] embeds an optimization problem in a policy and has been applied to HVAC control with a linear model [3], whereas DPC [5], adopted in this work, trains an explicit neural policy offline by differentiating a control objective through a model of the system. Learned world models [8,9] allow policies to be trained in imagination, at the risk of exploiting model errors [10]; Short-Horizon Actor-Critic (SHAC) [15] addresses long horizons with a learned value function, whereas the present work differentiates directly through horizons of up to 48 hours.

3 Methods

3.1 Problem Setting

At every 5-minute step t, the policy receives the observation o_t of a building with V zones and K air handlers, with the measurements of every zone, air handler and hot-water system, the comfort flags, the occupancy and the time, and returns the action a_t, made of the damper and reheat commands of every zone and of the run states and setpoints of every air handler and of the hot-water system. The step reward combines occupant comfort, energy cost and carbon emissions as follows:

$$r_t = w_p\, R_t - w_e\, E_t - w_c\, C_t, \tag{1}$$

where w_p, w_e and w_c are the weights fixed by the competition, E_t and C_t are the energy cost and the carbon emissions of electricity and natural gas, each normalized by a reference value, and $R_t \le 0$ is the productivity regret of the occupants, equal to zero when the building is unoccupied and otherwise defined as:

$$R_t = -\frac{p_{\max}}{p_{\max} - p_{\min}} \frac{\sum_v n_{v,t}\, \psi(\delta_{v,t})}{\sum_v n_{v,t}}, \qquad \delta_{v,t} = \max\left(T_{v,t} - \overline{T}_{v,t},\ \underline{T}_{v,t} - T_{v,t},\ 0\right), \tag{2}$$

where $n_{v,t}$ is the number of occupants of zone v, $T_{v,t}$, $\underline{T}_{v,t}$ and $\overline{T}_{v,t}$ are its air temperature and its heating and cooling setpoints, $p_{\max}$ and $p_{\min}$ are the maximum and minimum productivity per person-hour, and ψ is the productivity shortfall, equal to δ/δ_0 up to the reference deviation $\delta_0 = 5$ K and to $1 + \big((\delta - \delta_0)/\delta_0\big)^2$ beyond it, thereby growing linearly and then quadratically outside the comfort band. The score of a policy is the mean absolute step reward $S = \frac{1}{N}\sum_t |r_t|$ over the N steps of all scenarios, so that lower is better.

3.2 Entity-Token Architecture

As can be observed in Figure 1(b), a building is represented by $N_b = V + K + 2$ tokens, one per zone, one per air handler, one for the hot-water system and one global token. The input in_i of a zone concatenates its normalized observations, last three temperature changes, previous commands and seven static descriptors of the floor plan; the other tokens hold their sensors and previous commands, and the global token the comfort-mode flags, occupancy, time and, for the policy, the hours to the next comfort period. Every input vector is first embedded as follows:

$$x_i^{(0)} = \mathrm{MLP}_{c(i)}(\mathrm{in}_i) + e_{c(i)}, \tag{3}$$

where $c(i) \in \{\mathrm{zone}, \mathrm{ahu}, \mathrm{hws}, \mathrm{glob}\}$ is the type of token i, MLP_c is a two-layer Multi-Layer Perceptron (MLP) specific to that type and $e_c \in \mathbb{R}^d$ is a learned type embedding of dimension d, the model width. The tokens are then processed jointly by L pre-norm transformer layers [12], each made of a multi-head attention block and a feed-forward block with Gaussian Error Linear Unit (GELU) activation, both with residual connections, and the only structural information about the building enters through the attention weights of head h in layer ℓ, in the spirit of Graphormer [16]:

$$A_{ij}^{(\ell,h)} = \mathrm{softmax}_j\left(\frac{q_i^{(\ell,h)} \cdot k_j^{(\ell,h)}}{\sqrt{d/n_h}} + b_{\rho(i,j)}^{(\ell,h)}\right), \tag{4}$$

where $A_{ij}^{(\ell,h)}$ weights the value of token j in the update of token i, the query $q_i^{(\ell,h)}$ and the key $k_j^{(\ell,h)}$ are linear projections of the normalized tokens, n_h is the number of heads, $\rho(i,j)$ is one of ten relations between two tokens, determined only by the map of zones to air handlers (for instance, two zones on the same air handler, or a zone and its own air handler), and $b_\rho^{(\ell,h)}$ is a learned scalar per relation, layer and head, initialized to zero. Since no parameter depends on V or K, the same weights apply to every building. Linear heads read the output tokens $x_i^{(L)}$; the continuous commands of the policy pass through a hyperbolic tangent, and its on/off commands through a straight-through threshold.

3.3 World Model

Given the state s_t, that is, the token inputs above, the action a_t and the exogenous inputs ξ_t read from the episode (weather, setpoints, occupancy, tariffs and carbon intensities), the world model f_ϕ predicts the next actuator readbacks and sensor values of every entity, the zone temperatures and the power of the building, the last two as follows:

$$\hat{T}_{v,t+1} = \hat{T}_{v,t} + \sigma_{\Delta T}\, y_{v,t}^{T},$$
$$\hat{P}_t^{\mathrm{el}} = P_0^{\mathrm{el}}\Big[\sum_{g=1}^{K} \sigma_+\big(y_{g,t}^{\mathrm{el}}\big) + \sigma_+\big(y_{\mathrm{hws},t}^{\mathrm{el}}\big)\Big], \qquad \hat{P}_t^{\mathrm{gas}} = P_0^{\mathrm{gas}}\Big[\sum_{v=1}^{V} \sigma_+\big(y_{v,t}^{\mathrm{gas}}\big) + \sigma_+\big(y_{\mathrm{hws},t}^{\mathrm{gas}}\big)\Big], \tag{5}$$

where $y_{v,t}^{T}$ is the output of the zone head for the normalized temperature change, $\sigma_{\Delta T}$ the standard deviation of the 5-minute temperature changes in the training data, y^{el} and y^{gas} the outputs of the heads of air handler g, zone v and hot-water system for the electric power (fans, cooling, pumps) and the gas power (reheat, boiler), $P_0^{\mathrm{el}} = 80$ kW and $P_0^{\mathrm{gas}} = 205$ kW scale constants, and σ_+ the softplus function, which keeps every contribution non-negative, and the reward (1) is rebuilt exactly from $\hat{T}$ and $\hat{P}$. Since walls and interior mass store heat that is not observed, each zone token is augmented by a long memory of exponential averages:

$$m_{v,t+1}^{(z,\tau)} = \beta_\tau\, m_{v,t}^{(z,\tau)} + (1 - \beta_\tau)\, z_{v,t+1}, \qquad \beta_\tau = e^{-1/\tau}, \tag{6}$$

where $m_{v,t}^{(z,\tau)}$ is the exponential average of the signal z with time constant τ, and z is the zone temperature (τ of 1, 6 and 24 hours), the reheat and damper commands and the outdoor air temperature (6 and 24 hours); during a rollout the averages are updated from the predicted temperatures $\hat{T}$.

The model is trained on closed-loop unrolls of $U \in \{48, 144, 288\}$ steps (4, 12 and 24 hours), extended to 576 steps (48 hours) in the last round, in which its own predictions are fed back while the logged actions are replayed, by minimizing the following loss:

$$\mathcal{L}(\phi) = \frac{1}{U}\sum_{k=1}^{U}\Big(\frac{\sum_v \omega_{v,k}\,(\hat{T}_{v,k} - T_{v,k})^2}{\sigma_{\Delta T}^2 \sum_v \omega_{v,k}} + \mu_R\Big(\frac{\hat{R}_k - R_k}{\sigma_R}\Big)^2 + \mu_s\,\mathcal{L}_k^{\text{sens}} + \mu_P\,\big\|\hat{p}_k - p_k\big\|^2\Big), \quad (7)$$

where k counts the steps of the unroll, the weight $\omega_{v,k}$ equals 4 for occupied zones outside the band or less than 1 K from its edges and 1 otherwise, thereby emphasizing the states in which the reward is sensitive to temperature errors, the second term matches the regret (2) directly, with $\sigma_R = 0.1$, $\mathcal{L}_k^{\text{sens}}$ the error on normalized readbacks and sensors, p_k the power normalized by KP_0, and μ_R, μ_s, μ_P the weights in Table 1.

Table 1: Architecture and training settings of the world model and of the final policy.

World Model	
Layers, width, heads, feed-forward	4, 128, 4, 256
Parameters	0.61 M
Unroll lengths U [steps]	48, 144, 288, 576
Loss weights μ_R, μ_s, μ_P	5, 0.3, 10
Memory time constants τ [steps]	12, 72, 288
Batch, steps per round	16 unrolls, 1000–1600
Optimizer, learning rate	AdamW, $5 \cdot 10^{-5}$

Policy & DPC	
Layers, width, heads, feed-forward	3, 64, 4, 128
Parameters	0.12 M
Horizon H (curriculum) [steps]	144, 288
Discount γ, damping λ	0.995, 0.99
Comfort factor κ, filter α	1.8, 0.3
Batch B, iterations I	24 starts, 400
World models, weekend starts	WM-3 & WM-2, 30%
Optimizer, learning rate	AdamW, $3 \cdot 10^{-4}$

The training data come from about 140 buildings of the official generator, with seeds disjoint from the official scenarios, under six weather stations. The first episodes come from an exploration policy (PID baseline, noise, random setpoints and extreme plant states); afterwards, policy and world model alternate in on-policy rounds (Algorithm 1), which correct the world model in the states that the policy visits. Section 4 compares the three most recent world models, all with memory (6): WM-1 (after a round of a filtered policy), WM-2 (a further round with plant setpoints held at random levels) and WM-3 (fine-tuned from WM-1 on 48-hour unrolls), while earlier ones are used only for the first two stages of Figure 2.

Algorithm 1 On-policy rounds of world-model training and DPC training of the policy.

$\mathcal{D} \leftarrow$ exploration episodes on the generated buildings; train f_ϕ on $\mathcal{D}$ with (7)
for round $m = 1, 2, \ldots$ **do**
  for I iterations **do** ▷ DPC, warm-started from an earlier policy
    $H \leftarrow$ curriculum; draw a world model f_ϕ of the round; sample B starts t_0 of an episode; $\hat{s}_{t_0} \leftarrow s_{t_0}$; $\tilde{u} \leftarrow$ logged command
    for $k = 0, \ldots, H-1$ **do**
      $\tilde{u} \leftarrow \alpha\,\pi_\theta(\hat{s}_{t_0+k}) + (1-\alpha)\,\tilde{u}$ (10); $\hat{s}_{t_0+k+1} \leftarrow f_\phi(\hat{s}_{t_0+k}, \tilde{u}, \xi_{t_0+k})$; memory (6)
      $\hat{r}_{t_0+k} \leftarrow$ reward rebuilt from $\hat{T}$ and $\hat{P}$ (8); damp the state gradient (9)
    end for
    $\theta \leftarrow \mathrm{AdamW}(\theta, -\nabla_\theta J)$ with clipped gradient norm, unless J is not finite
  end for
  $\mathcal{D}_m \leftarrow$ one 14-day episode per generated building with π_θ, partly with action noise
  fine-tune f_ϕ on $\mathcal{D} \cup \mathcal{D}_m$, half of every batch from $\mathcal{D}_m$; $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_m$
end for

3.4 Differentiable Predictive Control

The policy is trained by DPC [5], as illustrated in Figure 1(c). Starting from true states s_{t_0} of the collected episodes, the policy and the world model are rolled out in closed loop, and the following objective is maximized by backpropagation:

$$J(\theta) = \mathbb{E}_{s_{t_0}}\Big[\frac{1}{H}\sum_{k=0}^{H-1}\gamma^k\,\hat{r}_{t_0+k}\Big], \qquad \hat{r}_t = \kappa\,w_p\hat{R}_t - w_e\hat{E}_t - w_c\hat{C}_t, \quad (8)$$

where H is the horizon, γ the discount factor, $\hat{r}_t$ the reward (1), with its comfort term scaled by $\kappa > 1$, computed from the predicted temperatures and power of (5); κ counteracts the tendency of the policy to trade comfort for energy inside the world model. Three mechanisms make the optimization over hundreds of steps stable and transferable to the simulator. First, the gradient carried by the predicted state from one step to the previous one is damped without changing its value:

$$\hat{s}_{k+1} \leftarrow \mathrm{sg}(\hat{s}_{k+1}) + \lambda\big(\hat{s}_{k+1} - \mathrm{sg}(\hat{s}_{k+1})\big), \tag{9}$$

where $\mathrm{sg}(\cdot)$ is the stop-gradient operator and $\lambda < 1$ the damping factor, so that the credit assigned to an action by a reward Δk steps later is weighted by $\lambda^{\Delta k}$, thereby limiting the growth of the gradient over long horizons. Second, since the policy otherwise reacts to world-model errors with fast oscillating commands, which do not pay off in the simulator, every continuous command is filtered both inside the rollouts and at deployment:

$$\tilde{u}_t = \alpha\, u_t + (1 - \alpha)\, \tilde{u}_{t-1}, \tag{10}$$

where u_t is the command produced by the policy, $\tilde{u}_t$ the command applied to the building and $\alpha \in (0, 1]$ the filter coefficient; on/off commands are not filtered, and the policy receives $\tilde{u}_{t-1}$ as its previous action, exactly as at deployment. The intermediate policies of Section 4 apply the filter only at deployment (Filter), the final one also inside the rollouts (Filter In Training). Third, the horizon grows from 12 to 24 hours (from 6 to 12 in the intermediate policies), long enough to experience preheating and the overnight drift. In the final policy, every DPC iteration draws one of two world models (WM-3 and WM-2), so that the policy cannot exploit the errors of a single model, the policy also receives the averages (6) of its zones and a holiday flag, and 30% of the start states fall on Fridays and Saturdays, to represent the weekend and the following start-up. Every policy after the first is warm-started from an entity-token policy trained on an earlier world model.

4 Results

Policies are evaluated in the simulator over 14 days, so as to include a weekend and the following Monday, on the 30 official scenarios (six buildings under five weather stations), which are not used for training but are used to select the policies, so that only the hidden Kaggle scenarios are an independent test; the ratio of the Kaggle score to this score was 1.23, 1.29, 1.23 and 1.32 for the four submitted policies (1.39 and 1.48 over 7 days). The contribution of each stage of the method is broken down in Figure 2 by reward term, over the 30 scenarios, and by climate.

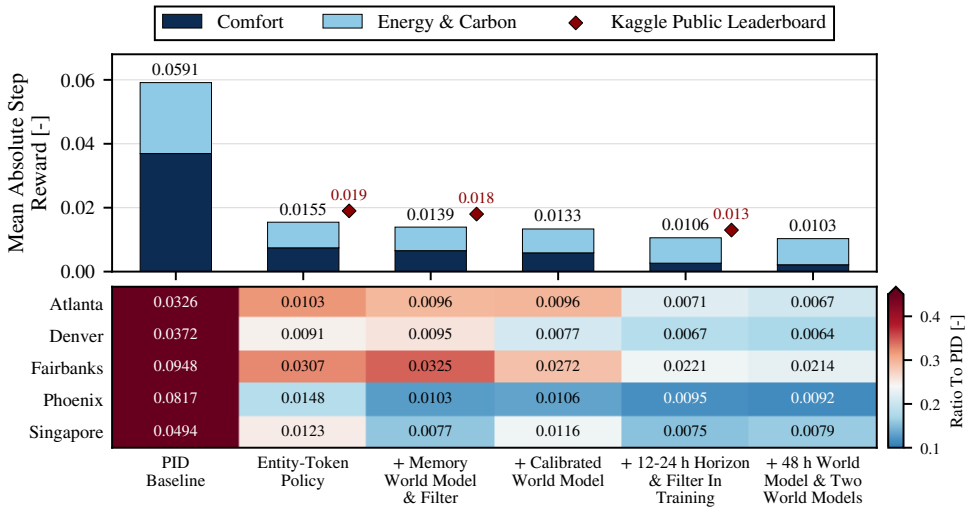


Figure 2: Score of the PID baseline and of the stages of the method, each adding the change in its label, on the 30 official scenarios over 14 days: reward terms and Kaggle scores (top), climate means colored by their ratio to the PID baseline (bottom).

4.1 Where the Gains Come From

As can be observed in Figure 2, the first entity-token policy already removes 74% of the PID score, reducing both terms in every climate. The following stages act on different terms and climates, since the world model with thermal memory and the command filter mainly benefit the warm climates (Singapore and Phoenix), the calibrated world model the cold ones (Fairbanks and Denver) at the expense of Singapore, and the 12 to 24-hour horizon with the filter inside the training rollouts more than halves the comfort term at almost equal energy, thereby improving every climate. Indeed, the filter alone, in rollouts of 6 to 12 hours, worsens the score (0.0182 against 0.0133). The final policy, trained on two alternated world models, lowers the comfort term further (to 0.0021) and reaches 0.0103 (-83%), while the six Fairbanks scenarios still account for about 40% of its loss.

4.2 Learned Behavior

Figure 3 compares the final policy with the PID baseline and the policy with memory and filter in Fairbanks 1 in January (outdoor air below $-35\,^{\circ}\mathrm{C}$) and Denver 8 in November.

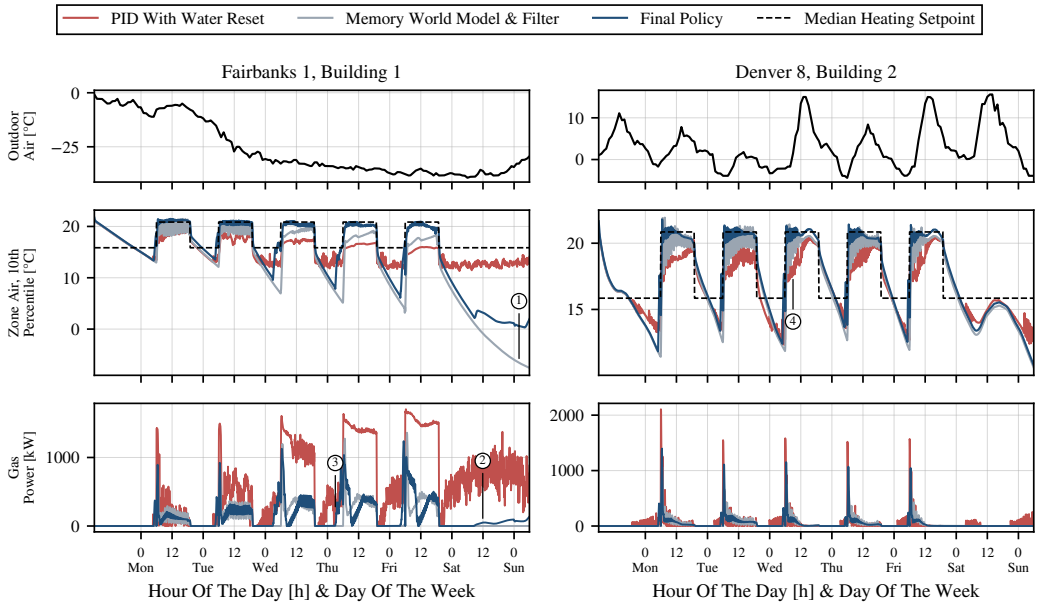


Figure 3: First week of Fairbanks 1 (from 1 January) and Denver 8 (from 5 November): outdoor air, 10th percentile of the zone temperatures with the median heating setpoint and gas power. The numbered markers are discussed in Section 4.2.

As shown, the PID baseline heats all night (marker 3), with 95.8 MWh of gas in the Fairbanks week against 23.9 MWh for the final policy, and still leaves the coldest zones below the setpoint, also in Denver 8 (marker 4). Both learned policies switch the plant off at night and over the weekend, but in Fairbanks 1 the policy with memory and filter lets the coldest zones fall below $-5\,^{\circ}\mathrm{C}$ (marker 1), whereas the final policy restarts heating on Saturday (marker 2) and keeps the coldest zones at the setpoint until the evening, with 21% more gas (score 0.0541 against 0.0650), and in Denver 8 it reaches the setpoint sooner (0.0121 against 0.0174).

4.3 World Models and Training Seeds

The three world models of Figure 4 reach similar errors over 24-hour closed-loop rollouts on held-out buildings (0.82, 0.79 and 0.79 K for WM-1, WM-2 and WM-3).

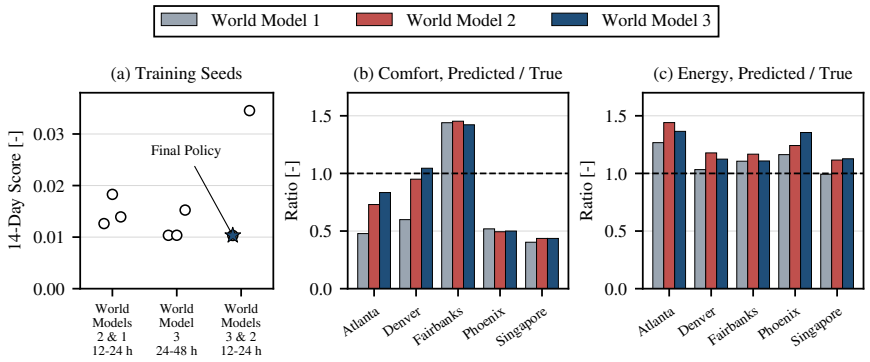


Figure 4: (a) 14-day score of every training seed; (b, c) predicted to true comfort and energy, by climate.

As can be observed in Figure 4(a), the training seed weighs as much as the choice of world model, since the recipe of the final policy scores 0.0103 with one seed and 0.0345 with another, and the recipe on WM-3 with a 24 to 48-hour horizon and $\gamma = 0.999$ yields 0.0104 in two of three seeds. Figures 4(b) and (c) show that all three world models underestimate the discomfort by about half in Phoenix and Singapore and overestimate it by more than 40% in Fairbanks, while they overestimate the energy, especially in Atlanta and Phoenix, so that their aggregate calibration on comfort (1.00 to 1.08) hides opposite errors across climates, which a single comfort factor κ cannot compensate. The policy on WM-2 is worse than that on WM-1 (0.0121 against 0.0106) in the warm climates, yet WM-3, with similar errors, yields the best policies, so that these world models differ less than the seeds.

4.4 Remaining Headroom

To bound the gain of better hourly decisions, an oracle clones the simulator every hour, evaluates 31 modifications of the next-hour commands of the final policy (plant off or on, water, supply-air and reheat offsets) over 24 hours and applies the best one, as compared in Figure 5 over 7 days with the same search scored by world models.

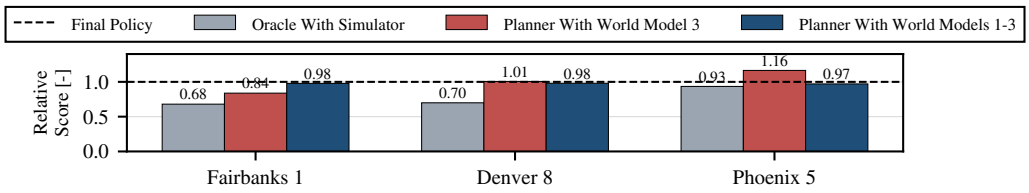


Figure 5: Seven-day score of the oracle and of the planners, relative to the final policy.

As can be observed, the oracle reduces the score by about 30% in Fairbanks 1 and Denver 8 but only marginally in Phoenix 5, while the search scored by WM-3 recovers half of the gap in Fairbanks 1 but worsens Phoenix 5, and the agreement of the three world models avoids the losses but gains little; each decision, moreover, takes 13 to 33 s against less than 1.5 s on Kaggle.

5 Conclusion

This work has shown that a single policy for unseen buildings can be trained by differentiating the reward through rollouts of an entity-token world model, reducing the PID score by 83%. Since the remaining loss is mostly subarctic and an oracle still recovers about 30% of it in two scenarios, future work will model the power of each zone coil and air handler, so that heat goes only where needed.

Acknowledgments. To be completed.

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